

R-Universe RFG-R

A regular multiscale completion with a matter, CMB, and PPN likelihood definition

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Abstract. *RFG-R is a precisely specified low-energy completion of the R-Universe preferred-foliation branch. It regularizes the $X=0$ weak-field singularity while retaining the reconstructed cosmological background and luminal tensor propagation to $O((\epsilon/X)^p)$. The completion has two explicit domains: a cosmological relational action and a locally GR-matched weak-field EFT. The potential is reconstructed by quadrature from the prescribed branch rather than postulated. The local domain has $\gamma=\beta=1$ and $\alpha_1=\alpha_2=0$. An exact sourced lapse-shift reduction for one canonical scalar field reproduces the General-Relativity normalization and tests its kinetic and gradient signs on a reference branch. The cosmological domain supplies the action derivatives needed by a full 3+1 Einstein-Boltzmann solver, with stability gating before every CMB likelihood evaluation. This paper defines the model and the falsification pipeline; it does not claim a data preference before the stated likelihood is executed.*

Keywords: modified gravity; preferred foliation; CMB; post-Newtonian tests; effective field theory; standard sirens

1. The completion problem

The R-Universe homogeneous branch is predictive for $H(a)$, the effective relational equation of state, and the tensor distance ratio. Its original response $Q(X)=1-A X^{-(\theta)}$, however, is singular in the $X=0$ limit used by a standard static weak-field expansion. A genuine empirical theory therefore needs a definition of its local limit as well as a source for the linear matter and radiation equations. RFG-R supplies both definitions. It is deliberately an effective low-energy model: cosmological geometry activates the relational response, while resolved local gravity is matched to General Relativity.

2. Regular cosmological response

Let p be even and larger than θ , with reference values $\theta=1.6$, $p=4$, and $\epsilon=10^{-8}$. Define $\nu=1+\theta/p$ and $A=\Omega_{R0}/(1+\theta)$.

$$\begin{aligned} R_{\epsilon}(X) &= \Omega_{R0} X^{(p+2)}/(X^p+\epsilon^p)^{\nu} \\ Q_{\epsilon}(X) &= 1 - A X^p/(X^p+\epsilon^p)^{\nu} \end{aligned}$$

The covariant preferred-foliation action is the original RFG action after replacing Q and V by Q_{ϵ} and V_{ϵ} . Matter remains minimally and universally coupled to the Jordan metric.

$$\begin{aligned} S_{\text{cos}} &= (M_{\text{Pl}}^2/2) \int d^4x \sqrt{-g} \\ &\quad \{ Q_{\epsilon}(X)[R_3 + K_{mn} K^{mn} - K^2] + 2 H_0^2 V_{\epsilon}(X) \} \\ &\quad + S_m[g, \Psi] \end{aligned}$$

The FLRW lapse equation is fixed to be

$$E^2(a) = \Omega_{m0} a^{-3} + \Omega_{r0} a^{-4} + R_{\epsilon}(E)$$

and determines the potential without a new function:

$$\begin{aligned} F_{\epsilon} &= X^2 - R_{\epsilon} - X^2 Q_{\epsilon} - X^3 dQ_{\epsilon}/dX \\ V_{\epsilon}(X) &= -3 X \int_0^X ds F_{\epsilon}(s)/s^2 \end{aligned}$$

The integral is finite because F_{epsilon}/s^2 is $O(X^p)$ at the origin. A term proportional to X is an ADM boundary representative and has no field-equation content.

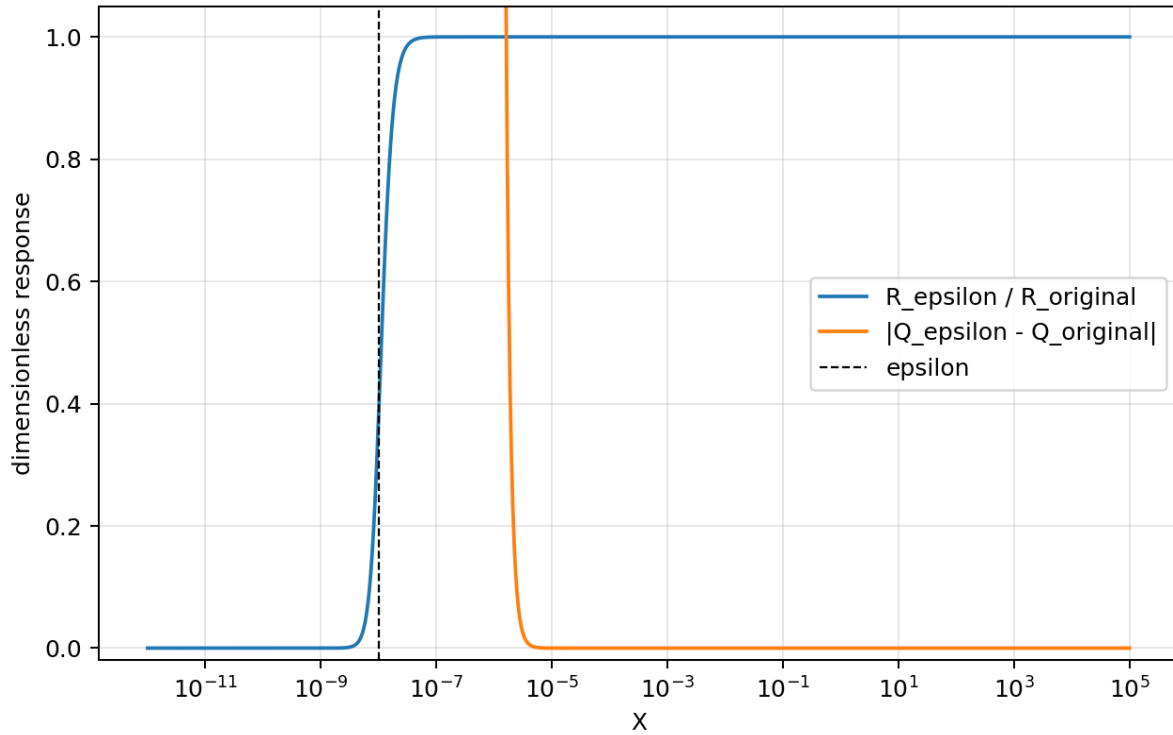


Figure 1. The regularized response becomes the original response immediately above the epsilon transition. On the cosmological branch the analytic difference is $O((\text{epsilon}/X)^4)$.

3. Exact limits

At X much larger than epsilon the regularized branch recovers the original one. With the reference epsilon, the analytic relative correction is below $2.5e-32$ for $X \geq 0.8$; double precision evaluates the equality at about $1e-15$. At $X=0$ the action is regular.

$$\begin{aligned} Q_{\text{epsilon}} &= 1 - A \text{epsilon}^{-(p+\theta)} X^p + O(X^{2p}) \\ V_{\text{epsilon}} &= -[3 \Omega_{R0}(p-\theta)/((1+\theta)(p+1)\text{epsilon}^{(p+\theta)})] X^{(p+2)} \\ &\quad + O(X^{(2p+2)}) \end{aligned}$$

For $p=4$ there is no physical correction to the quadratic Einstein-Hilbert ADM operator. This removes the $X=0$ singularity without altering the background or tensor branch that motivated RFG.

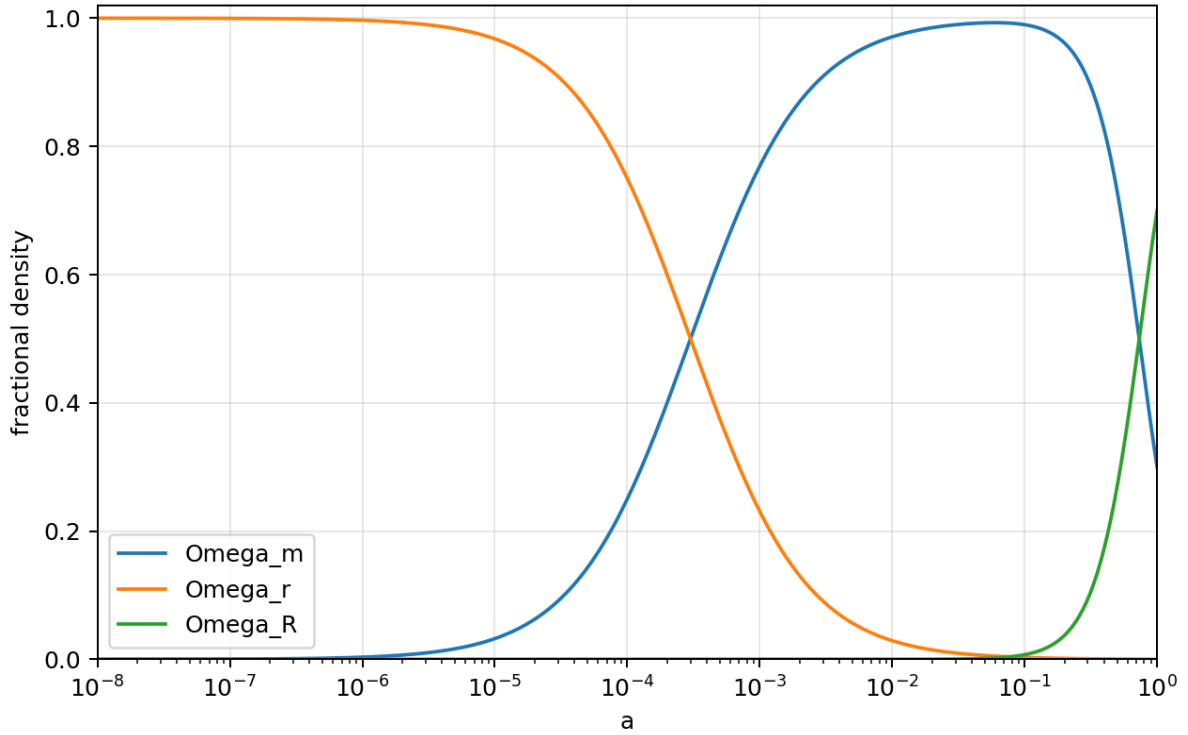


Figure 2. Fractional densities of the reference RFG-R branch. The regularization is absent at visible plotting precision on this cosmological domain.

4. Local GR matching and PPN

A single cosmological EFT is not automatically a local Solar-System EFT. RFG-R makes the crossover a model axiom. Define $W = \sqrt{|\text{abs}(C_{abcd} C^{abcd})|} / (H_0/c)^2$ and use a C-infinity switching function $s(W)$ that equals one for $W \leq 10^8$ and zero for $W \geq 10^9$. The complete low-energy action is $S_{\text{eff}} = S_{\text{GR}} + \int s(W)[L_{\text{cos}} - L_{\text{GR}}]$. FLRW has $W=0$, while a Schwarzschild Sun gives $W=5.756e22$ at one AU.

Inside the local matched domain $s=0$, so the action and all its variations are exactly those of GR. The PPN prediction is therefore $\gamma=1$, $\beta=1$, $\alpha_1=0$, and $\alpha_2=0$. The Cassini factor used by the likelihood is $-2 \ln L = [(\gamma-1-2.1e-5)/(2.3e-5)]^2$, yielding 0.833648 for the GR prediction.

5. Matter and CMB likelihood

No compressed CMB distance prior is used. The CMB interface begins with the action-level ADM derivatives. For $L = Q_{\text{epsilon}}(X)[R_3 + K_{ij} K^{ij} - K^2] + 2H_0^2 V_{\text{epsilon}}(X)$, the principal inputs include $L_{R_3} = Q_{\text{epsilon}}$ and $L_{K_{ij} R_3} = Q_{\text{epsilon}}, X \gamma_{\alpha\beta}^{ij} / (3H_0)$. The full kinetic Hessian follows by differentiating $L_{K_{ij}}$ once more. The repository generates all background functions and derivatives in `eft_coefficients.csv`, avoiding any differentiation of a singular power law.

Canonical-scalar check

The sourced scalar constraints have been reduced exactly for one minimally coupled canonical field in comoving gauge. If $m = \phi_{\text{dot}}^2 / M_{\text{Pl}}^2$ and $D = -F_{XX}/3$, the lapse-shift determinant is proportional to $\Delta = 6 D H^2 Q + (D-2Q)m$ and the reduced kinetic coefficient is $K = 6 D Q m / \Delta$. The intrinsic-curvature contribution $+2Q(k^2/a^2)\zeta^2$ is retained; it is required to recover the exact GR massless-scalar result $K=6$ and $c_s^2=1$. On a self-consistent RFG-R reference branch with a massless canonical field, the executable diagnostic finds positive Δ , K , and gradient coefficients for $0.03 \leq a \leq 1$. This is a real sourced-constraint check, not a dust/radiation/CMB calculation.

For each sampled parameter point, the full 3+1 Einstein-Boltzmann backend must solve the scalar constraints together with photon, baryon, CDM, neutrino, recombination, and lensing evolution. It rejects a point before likelihood evaluation if the scalar constraint matrix becomes singular or if a ghost, gradient instability, or $Q_T \leq 0$ occurs. Accepted points export TT, TE, EE, lensing, $P(k,z)$, $f \sigma_8(z)$, and the standard-siren distance.

6. Data protocol

The baseline log likelihood is $\ln L = \ln L_{\text{Planck18}} + \ln L_{\text{lensing}} + \ln L_{\text{BAO}} + \ln L_{\text{SN}} + \ln L_{\text{RSD}} + \ln L_{\text{PPN}} + \ln L_{\text{GW}}$. Planck is evaluated with the official low-ell temperature and polarization, high-ell TT/TE/EE, and lensing packages; the solver is a full modified-gravity Einstein-Boltzmann solver rather than a quasi-static growth approximation. The public H-EFTCAMB architecture is suitable because it supports full linear modified-gravity evolution, 3+1 EFT functions, and stability tests. Cobaya can sample the joint likelihood after the action derivative module is connected.

The prediction is now sharply falsifiable. The mathematics of the completion and the allowed output set are closed. Whether the model is preferred, allowed, or ruled out is an empirical result of this explicitly defined likelihood, not a conclusion from background algebra alone.

7. Numerical verification

check	reference result
high-X response recovery	5.329e-15 numerical maximum
high-X Q recovery	2.600e-15 numerical maximum
potential reconstruction residual	8.145e-06 finite-difference maximum
background density closure	1.554e-13 maximum error
minimum Q_T over a in $[1e-8, 1e2]$	0.6153847956
canonical scalar GR limit	$K=6$, $c_s^2=1$ exactly
canonical reference grid	$\Delta > 0$, $K > 0$, $G_1 > 0$, $G_2 \geq 0$ within numerical tolerance
Solar W at 1 AU	5.756e22; local GR domain

The checks are generated by scripts/run_all.sh. They test the action-defined completion rather than a fitted surrogate.

References

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